Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Engineer with 8+ years of production software engineering and hands-on generative AI development. I build LLM-powered applications end-to-end: LeaseI is an 8-agent LangGraph pipeline with Retrieval-Augmented Generation (RAG), multimodal document intelligence, and HITL escalation (93.6% precision on 1,240 lease PDFs); AgentProbe is a production eval harness with a 15-endpoint FastAPI backend, 5-metric LLM-as-judge engine, vector database (ChromaDB) integration, and CI/CD regression gate. Strong hands-on experience with LangChain, LangGraph, RAG pipelines, prompt engineering, function calling, and AI agent frameworks across OpenAI, Anthropic, Groq, and Ollama APIs. Production backend depth: 8 years Python at scale (100K+ MAU, 1M+ records, 120+ Agile sprints), FastAPI REST APIs, Docker, AWS, PostgreSQL, and structured + unstructured data pipelines. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

LangChain & LangGraph | Retrieval-Augmented Generation (RAG) | Large Language Models & Generative AI | FastAPI & REST API Design | Vector Databases (ChromaDB, Pinecone) | Prompt Engineering & Function Calling | Machine Learning & Deep Learning | Production AI Deployment

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building machine learning models and agentic AI pipelines; integrating structured and unstructured data systems with back-end APIs; iterating on production AI application behavior and model performance.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern

- Built an LLM-powered agentic application using LangGraph (parse → plan → retrieve → validate → conditional routing) delivering structured, compliant outputs from natural-language inputs -- integrated with enterprise systems via REST APIs.
- Implemented a RAG pipeline (retrieval over FAA Part 107 corpus) with prompt engineering and function calling for grounded compliance decisions; deployed with Docker; engineered AI guardrails blocking non-compliant outputs before reaching users.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Fine-tuned a custom deep learning classification model (HuggingFace Transformers + PyTorch); deployed inference API on Quart; evaluated model performance and presented results to faculty stakeholders.
- Built multi-source machine learning prediction pipelines (Python, R, $R^2 = 0.977$); applied Isolation Forest for automated anomaly detection; designed Tableau/Quarto dashboards (+40% comprehension).

Entrata (PropTech / Real Estate SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Shipped 20+ production Python modules** across 120+ Agile sprints -- REST APIs, async pipelines (RabbitMQ), microservices (Docker, 50% scalability improvement), and database-backed services for 100K+ monthly active users at 99.5% uptime. Integrated AI services and third-party APIs across enterprise systems end-to-end.
- **Built and owned data pipelines for structured and unstructured data** -- annual income-limit imports (few thousand source rows → 180,000 domain records), eligibility and financial calculations, AWS S3 document migration with Libsodium encryption (100% PHI compliance, zero incidents).
- Optimized 200+ SQL queries (30% performance gain); eliminated 80% of manual operational workload via Python/Bash automation pipelines; mentored 5+ engineers; drove technical standards via 300+ code reviews.

PROJECTS

AgentProbe: Open-Source AI Agent Eval Harness

LangGraph -- LangChain -- FastAPI -- ChromaDB -- CI/CD

Production eval harness for LLM-powered applications and AI agents: 15-endpoint FastAPI REST API backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), ChromaDB vector database integration, CI/CD regression gate automatically blocking failing agent runs, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), prompt engineering pipeline with version management, observability dashboard. Deployed with Docker; used to evaluate LeaseI across 1,240+ compliance test cases. Supports function calling, tool use, and conversational AI evaluation.

github.com/venkatalakshmiparimala/AgentProbe

Leasel: Agentic RAG Compliance Investigation Platform LangGraph 8-agent -- RAG -- FastAPI -- PyTorch -- Multimodal

LLM-powered enterprise AI application built on LangGraph: 8-agent stateful pipeline (intent classification → multi-source retrieval → deep learning reasoning → HITL escalation) over structured and unstructured data (lease PDFs, system configs, regulatory rules, historical incidents). RAG implementation: multimodal OCR ingestion, TF-IDF + semantic hybrid search, Pinecone-compatible vector retrieval, citation grounding, confidence scoring. REST APIs via FastAPI; deployed on AWS-compatible infrastructure; CI/CD via GitHub Actions. Results: 93.6% precision / 87.9% F1 on 1,240 lease PDFs; 100% citation correctness on Phase 1 eval set.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AI on FHIR: NLP-Driven Clinical Records Query System Flask -- LLMs -- BERT -- NLP -- React -- Docker

LLM-powered NLP application translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT embeddings, and deep learning; HIPAA-compliant RBAC across structured medical records; reduced manual clinical retrieval time ~60%; deployed with Docker. Demonstrates end-to-end AI integration with enterprise systems and third-party APIs.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur 2017

SKILLS

AI & Agents: LangChain, LangGraph, Large Language Models (LLMs), Generative AI, RAG Pipelines, Retrieval-Augmented Generation, Vector Databases, ChromaDB, Pinecone, FAISS, Prompt Engineering, Function Calling, AI Agents, Conversational AI, Multi-Agent Orchestration, HITL, AI Guardrails, Hallucination Detection, OpenAI API, Anthropic API, Groq, Ollama, Hugging Face Transformers, LlamaIndex, LLMOps, MCP

ML & Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, Isolation Forest, NLP, Deep Learning, Model Training, Model Evaluation, Model Optimization, Model Inference, Feature Engineering, Pandas, NumPy, MLflow

Languages & Frameworks: Python, JavaScript, TypeScript, SQL, NoSQL, FastAPI, Flask, REST APIs, React.js, Django, Spring Boot, Java, Node.js

Cloud & Infra: AWS (S3, EC2, Lambda, Glue, Bedrock), Azure, Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD (GitHub Actions), PostgreSQL, RabbitMQ, Redis, Microservices, Distributed Systems